

The Urban Wage Premium in Historical Perspective^{*}

Kyle Butts[†] Taylor Jaworski[‡] Carl T. Kitchens[§]

June 12, 2023

Abstract

We estimate the urban wage premium in the United States from 1940 to 2010. Drawing on recent advances in the literature on selection on unobservables, we show how to control for heterogeneity in the characteristics of individuals that choose to live in cities to address endogenous sorting. Estimates from naive comparisons of individuals living in urban versus rural areas substantially overstate the urban wage premium. We find that the premium is highest in the middle of the twentieth century (about 12 percent in 1940 and 1950) relative to the early in twenty-first century (declining to a few percent by mid-2020). Overall, the urban wage premium is decreasing and sorting explains a larger fraction of the difference in urban versus rural earnings across our sample period.

^{*}This version of the paper includes preliminary results. Please do not cite without permission.

[†]CU-Boulder, Department of Economics, (kyle.butts@colorado.edu)

[‡]CU-Boulder, Department of Economics, and NBER (taylor.jaworski@colorado.edu)

[§]Florida State, Department of Economics, and NBER (ckitchens@fsu.edu)

1 Introduction

Average wages in cities are higher than in nonurban areas across the globe. Within the United States, there is substantial evidence for this “urban wage premium.” For example, today the observed difference in wages between urban workers and their non-urban counterparts is more than 35 percent.¹ Going back in time, recent work shows that the difference between urban and rural wages has roughly doubled since 1980 (Dauth, Find-
eisen, Moretti and Suedekum, 2022). This striking pattern has motivated a large literature in urban economics to understand the drivers of the urban wage gap. In particular, is the difference driven by increased productivity stemming from agglomeration economies or sorting based on ability and correlated preferences (e.g. Behrens, Duranton and Robert-Nicoud, 2014; Moretti, 2011; Puga, 2010)? Understanding how the underlying drivers of selection and strength of agglomeration forces have varied over time is important for predicting the success of place-based policies (e.g. ?) or to understand growth trajectories in the context of endogenous growth models (e.g. ?).

The fact that wages tend to be higher in cities cannot be interpreted as the causal effect of cities due to the potential for non-random sorting of people or firms into cities based on preferences, skills, or productivity.² In particular, the key identification challenge arises in disentangling the productivity effect of cities from unobserved characteristics of individuals, firms, or cities induced by endogenous sorting.

Research in this area has typically addressed these concerns using detailed worker- and firm-level panel data. Comparing the same individuals before and after moving removes time-invariant wage variation due to worker characteristics, such as ability. Then, the remaining gap or subsequent wage growth between urban and rural earnings is attributed to agglomeration forces stemming from knowledge sharing, firm-worker matching, human capital accumulation, labor market pooling, or input-output linkages (Baum-Snow and Pavan,

¹ We use the 2010 ACS estimates to compute the average annual wage for individuals living in MSAs and outside of MSAs based on individuals that are in the labor force, report non-negative wage income, and who are not missing wage income data.

² For example, positive selection on the firm side may come from more productive firms paying higher wages, and only high-productivity firms locating and surviving in cities where larger markets induce tougher competition.

2012; Combes, Duranton and Gobillon, 2008; Dauth, Findeisen, Moretti and Suedekum, 2022; Glaeser and Maré, 2001; Gould, 2007; Yankow, 2006). The previous literature provides a range of estimates regarding the urban wage premium. For example, Glaeser and Maré (2001) estimate an urban wage premium between 4.5 and 11 percent after controlling for individual characteristics and, more importantly, unobserved heterogeneity with person-specific fixed effects. Gould (2007) finds a premium of 11.5 percent for white-collar workers and a smaller premium of 1.2 percent for blue-collar workers.³

Due to limited data availability the existing literature has focused primarily on the last three decades and little is known about the evolution of the urban wage premium before 1980. Given the role of agglomeration in economic models of cities, regions, and national growth, it is important to understand long run changes in the causal effect of cities on wages. For the United States, Boustan, Bunten and Hearey (2013) provide descriptive evidence going back more than a century. Their key finding is a U-shaped pattern since 1940: the observed difference between urban and nonurban wages is approximately 35 percent in 1940, decreases to between 20 and 25 percent until 1980, and has since returned to be above 35 percent.

In our paper, we show how to use readily available data to estimate the urban wage premium while controlling for observed individual characteristics as well as unobserved characteristics following the approach in Altonji and Mansfield (2018). In their setup, the classic urban location choice model gives rise to a set of sufficient statistics that control for sorting based on unobserved characteristics. Importantly, this approach allows us to address sorting in the context of estimating the historical urban wage premium in the absence of individual-level longitudinal data. Specifically, we use only repeated cross-sections from the US decennial censuses and American Community Survey beginning in 1940 when information on individual wages was first reported.

We estimate the urban wage premium from 1940 to 2010. Focusing first on the period since 1980, where we can make direct comparisons to the existing literature, we document an urban wage premium of up to 5 percent in each decade. This is consistent with the results

³ Rosenthal and Strange (2004) and, more recently, Diamond and Gaubert (2021) survey the large literature that describes the theoretical underpinnings and empirical findings related to agglomeration and urban sorting.

reported in Glaeser and Maré (2001), Yankow (2006), and Baum-Snow and Pavan (2012) and provides support that our approach (drawing on Altonji and Mansfield (2018)) addresses the key identification concern related to sorting. Second, moving to earlier decades, we estimate an urban wage premium of 12.7 percent in 1940, 13.2 percent in 1950, 10 percent in 1960, and 5.6 percent in 1970. The pattern suggests that agglomeration forces were relatively more important prior to 1980, although sorting still drives a large portion of the observed premium between 1940 and 1980.⁴

The increasing role of sorting for explaining the difference between urban and nonurban wages over time is consistent with a large literature that highlights increased migration by highly educated workers to cities (Shapiro, 2006; Moretti, 2013; Diamond, 2016), changes in the structure of manufacturing after 1950 (Dumais, Ellison and Glaeser, 2002; Glaeser and Ponzetto, 2010), the increased mobility of highly skilled workers in response to tax policy (Moretti and Wilson, 2017; Verginer and Riccaboni, 2021; Akcigit, Baslandze and Stantcheva, 2016), and recent evidence of skilled workers sorting on improved local amenities (Baum-Snow and Hartley, 2020; Diamond, 2016; Su, 2022; Glaeser and Shapiro, 2003; Glaeser and Gottlieb, 2009).⁵

In addition to the main findings, we explore several potential sources of heterogeneity. First, we consider regional differences and confirm that urban wage premium is falling *within* each region. This suggests that it is not changes in regional development (i.e., the deindustrialization of the North and rise of the Sunbelt) driving changes in the urban wage premium. Second, we examine changes in the urban wage premium across the distribution of city sizes. We show that in 1940, the largest cities had stronger agglomeration forces relative to smaller urban areas; however, since the 1980s, the gap has narrowed. The pattern we document

⁴ These results are robust to alternate definitions of cities. In particular, due to changes in the definition of cities over time, we also estimate the urban wage premium on consistent set of locations based on the urban status in 1940.

⁵ Our results are also consistent with a growing literature that emphasizes the potential to overstate agglomeration forces in small samples (Ahlfeldt, Albers and Behrens, 2022; Dingel and Tintelnot, 2020; Schoefer and Ziv, 2022). For example, Schoefer and Ziv (2022) note that up to three-quarters of the variation in regional productivity dispersion is due to small sample bias. Similarly, Dingel and Tintelnot (2020) demonstrate that the potential impacts of Amazon’s HQ2 are smaller when accounting for granularity relative to the traditional spatial equilibrium models.

is consistent with the transition of cities to idea-generating rather than heavy manufacturing. Finally, we show that the urban wage premium is higher for college-educated workers. Despite the decline in the share of the premium attributable to agglomeration over time, college-educated workers continue to earn an urban premium. This is consistent with the role of skills in cities emphasized by [Moretti \(2013\)](#).

2 Data and Summary Statistics

In this section, we describe the data we use to estimate the urban wage premium and key empirical patterns regarding the changes in urbanization, education, and the urban-nonurban wage difference in the second half of the twentieth century. First, we draw on publicly available data covering the period between 1940 and 2010 ([Ruggles et al., 2022a,b](#)). For 1940, we use the complete count of the US Census to measure the average earnings per worker for wage earners. For 1960 and 1970, we use the 5 percent samples of the census. For 1950, 1980, 1990, and 2000, we draw on the 1 percent samples of the census. Finally, for 2010, we use the American Community Survey. In each sample, we restrict the data to include adult males living in the continental US between the ages of 25 and 65 who report being employed for at least 26 weeks in the sample year. We use person weights to calculate summary statistics and estimate regressions.

To construct average weekly earnings, we divide the reported total income by the number of weeks worked in the survey year.⁶ We then convert all earnings to 2015 dollars using the Consumer Price Index. In addition, across all samples, consistent information is available on whether the individual is living in an urban area (metropolitan statistical area), age, educational attainment, and race. We use this information to construct the individual- and group-level controls for our empirical analysis. We assign workers to a city based on the metropolitan statistical area (MSA) in which they reside; workers not living in a city are assigned to their state of residence.

⁶ For individuals earning the top income code, we set their total income to the lower top income cutoff multiplied by 1.5. These cutoffs are reported in the IPUMS documentation for the INCWAGE variable. In 2015 dollars, the cutoffs are \$84,500 for 1940, \$99,200 for 1950, \$199,500 for 1960, \$309,000 for 1970, and \$225,000 for 1980. Afterward, IPUMS reports income as the median wage above that year's cutoff. Results are robust to dropping all individuals earning the top income code.

We begin by describing a few key empirical patterns in Table 1. First, we report the raw observed difference between urban and rural wages in column 2. Here we document a clear U-shape in the summary statistics. The urban wage gap started at 31.3 percent in 1940 and declined sharply through 1980. Beginning in 1980, the urban wage gap increased until 2000 where it reached nearly the same level as reported in 1940. Next, in column 3, we report the share of the population living in urban locations each year. In 1940, 66 percent of individuals lived in urban areas. By 2010, the urban share increased to 79.6 percent. The pattern of increasing urbanization is consistent with prior literature such as [Moretti \(2013\)](#) and [Boustan et al. \(2013\)](#). To highlight the potential role of sorting, the remaining columns provide information the overall fraction of the population with college degree in each year and differences in this share across urban and nonurban areas. From column 4, the share of the population with a college degree increased over time. From columns 5 and 6, the share of the population with a college degree increases in both urban and rural areas over time; however, the share rises faster in cities. Together, the trends in Table 1 highlight that since 1940 cities have become more populated and more educated relative to rural areas, which suggests increased sorting based on skills. The goal of empirical approach to address the potential role of changes in the extent of sorting based on education, other demographic characteristics, and unobserved productivity over time.

The geography of high-wage urban centers has also changed over time. Table 2 reports the ten urban areas with the highest urban wage differences in 1940 and 2010. In particular, we calculate the difference in the average log wage in an urban and nonurban area. While several of the cities are among the highest paying in both 1940 and in 2010 (i.e., San Francisco, Washington, D.C., San Jose, Seattle), centers of manufacturing in the industrial North and Midwest (i.e., Flint, MI, Detroit, MI, Lansing, MI, Milwaukee, WI, Rochester, NY) have lost their prominence in favor of cities closely associated with high-tech manufacturing. This descriptive pattern reflects the combination of several changes in the US economy since 1940. The main goal of remainder of the paper is to quantify and better understand changes in the urban wage premium over time and the relative importance of agglomeration (i.e., the causal effect of cities) versus sorting.

Table 1: Summary Statistics

Year	% Wage Gap	% Urban	% With College Degree		
			Overall	in Urban Areas	in Nonurban Areas
(1)	(2)	(3)	(4)	(5)	(6)
1940	31.3	66.0	8.4	8.8	7.6
1950	21.9	68.7	8.0	8.6	6.7
1960	23.6	65.6	11.6	12.9	9.1
1970	23.6	74.7	16.6	17.8	13.0
1980	19.9	72.8	24.2	26.6	17.5
1990	25.6	73.5	28.0	31.1	19.6
2000	31.4	78.7	30.5	33.4	19.8
2010	27.8	79.6	34.1	37.0	23.0

Notes. This table presents summary statistics for the final ACS sample. The data cleaning process is described in Section 2.

3 Measuring the Urban Wage Premium

We begin by estimating the urban-rural wage difference in a simple regression framework. We follow the literature by adding individual-level covariates to address sorting into urban locations. In particular, we estimate the following equation separately for each year:

$$\log wage_i = \beta_0 + \beta_1 urban_i + X_i\delta + \varepsilon_i, \quad (1)$$

where $wage_i$ is average weekly earnings and X_i is a set of controls consisting of indicator variables for five-year age groups, educational attainment, and race (white versus non-white). Our estimate of β_1 is the average premium in percentage terms that urban workers earn relative to nonurban workers (we use percent by calculating $e^{\hat{\beta}_1} - 1$ in figures.) Without controls, the coefficient reflects the mean difference in earnings between individuals living in urban and nonurban locations. If the set of control variables fully accounts for sorting, we could interpret β_1 as the causal effect of urban locations on earnings. However, there are several reasons individuals sort into cities that are not captured by their observable characteristics. For example, individuals may sort to cities because they value specific consumptive ameni-

Table 2: Top Ten Cities by Urban Wage Premium in 1940 and 2010

1940		2010	
Name	Premium	Name	Premium
Flint, MI	48.7%	San Jose-Sunnyvale-Santa Clara, CA	63.2%
Detroit, MI	46.2%	Bridgeport-Stamford-Norwalk, CT	54.7%
San Francisco, CA	43.8%	San Francisco-San Mateo-Redwood City, CA	52.0%
Seattle-Everett, WA	43.0%	Washington-Arlington-Alexandria DC-VA	42.9%
Washington DC, MD/VA/WV	42.6%	Boston-Quincy, MA	42.7%
Lansing-East Lansing, MI	39.2%	Seattle-Bellevue-Everett, WA	42.2%
Sacramento, CA	39.0%	Santa Cruz-Watsonville, CA	36.7%
New York, NY	38.8%	Trenton-Ewing, NJ	36.1%
Milwaukee-Waukesha, WI	38.5%	Midland, TX	34.0%
Rochester, NY	38.4%	Baltimore-Towson, MD	33.6%

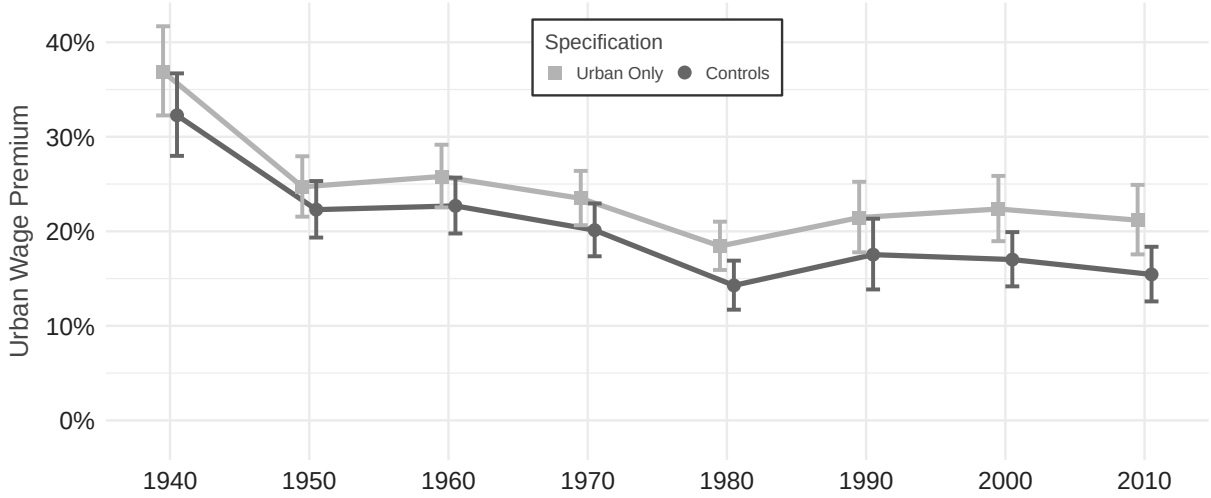
Notes. This table presents the top ten cities in terms of observed wage premia relative to non-urban areas.

ties, such as entertainment or dining. As a result, the estimated β_1 should be interpreted as a correlation.

The results of estimating equation (1) are displayed in Figure 1. The grey line with square markers reports the mean urban wage difference. The mean urban differential was highest in 1940 at around 35 percent and falls to a little under 25 percent by 1970. Starting around 1980, the urban wage difference stabilizes at around 20 percent, where it remains to the present. The medium grey line with circle markers reports estimates that include individual-level controls in the specification. Adding individual-level covariates reduces the wage differential in all years. In 1940, individual characteristics explained roughly 10 percent of the earnings gap between non-urban and urban locations. By 2010, differences in individual-level observables explained about 25 percent of the gap. Relative to prior literature such as Yankow (2006) and Glaeser and Maré (2001), our set of observables explains less of the wage differential; however, we are more limited in our set of observable individual-level characteristics.

Overall, the pattern of wage differentials is similar once individual-level characteristics are included. We also show that individual characteristics explain a larger share of the urban wage differential over time. This suggests that sorting – despite our limited set of observables

Figure 1: The Urban Wage Premium Controlling for Individual Characteristics, 1940-2010



Notes: This figure plots the coefficient for survey-year level estimates of the urban wage premium from equation (1) with different level of controls. The light gray line with square markers recreates the unadjusted point estimates. The dark gray line with circle markers includes controls for age in 5-year bins, education levels, and an indicator for being White. The points for each line are plotted with the 95 percent confidence intervals.

– is increasing over time and is consistent with the raw data in Table 1.⁷

4 Sorting and the Urban Wage Premium

In this section, we move on to quantifying the urban wage premium by controlling for unobserved characteristics that may drive sorting. The urban wage premium literature often uses individual panel data to isolate the effect of living in an urban area by observing individuals who transition across urban and rural locations.⁸ Given the cross-sectional nature of our data, we cannot adopt these panel data methods. To overcome the empirical challenge of addressing selection in the absence of individual-level panel data, we use properties of a location choice model to control for sorting as proposed by [Altonji and Mansfield \(2018\)](#).

Individuals with different characteristics sort to different locations based on location-

⁷ The reason for the difference in estimates comes from our use of a regression framework. The method in [Boustan et al. \(2013\)](#) estimates $\log(\hat{E}[\text{Wage}|\text{Urban} = 1]) - \log(\hat{E}[\text{Wage}|\text{Urban} = 0])$, which differs from the regression estimate, $\hat{\beta}_1 = \hat{E}[\log(\text{Wage})|\text{Urban} = 1] - \hat{E}[\log(\text{Wage})|\text{Urban} = 0]$. The latter is a more precise estimate for average percent difference in wages.

⁸ For a recent example that uses matched employee-employer data, see [Frings and Kamb \(2022\)](#). Additionally, ? utilizes the placement of refugees in Denmark to estimate the returns to urban experience.

specific amenities. This is apparent in the data based on differences in location decisions across individuals with different observable characteristics. The key insight from the approach in [Altonji and Mansfield \(2018\)](#) is that individuals also sort based on unobservables for the same amenities. Thus, observing differences in the average population characteristics reveals information about sorting on unobservables. [Altonji and Mansfield \(2018\)](#) show that in a setting where the indirect utility function is additively separable, the mapping from location-level averages of observable variables to averages of unobservable variables is linear.

The key insight from the approach in [Altonji and Mansfield \(2018\)](#) is that as more people choose a given location, the average characteristics of the residents differ across location choices only because they value location-specific amenities differently. Thus, observing differences in the average population characteristics reveals information about sorting on unobservables. In a setting where the indirect utility function is additively separable, the mapping from location-level averages of observable variables to averages of unobservable variables is linear. Following [Altonji and Mansfield \(2018\)](#), we utilize this mapping in our empirical approach by adding group-level averages of individual characteristics to our baseline estimating equation to control for sorting on unobservable characteristics.⁹

More formally, consider an indirect utility function over locations s that is additively separable in amenities A_s and prices P_s :

$$V_i(s) = W_i A_s + \varepsilon_{si} - P_s. \quad (2)$$

W_i represents individuals' underlying weights on each latent amenity. [Altonji and Mansfield \(2018\)](#) then define the weights, W_i , as a function of observable characteristics X_i , unobservable characteristics X_i^U .¹⁰ Under these assumptions, the utility function can be rewritten as a function of individual-level observables, individual-level unobservables, amenities, and prices.

$$V_i(s) = (X_i \Theta + X_i^U \Theta^U) A_s + \varepsilon_{si} - P_s. \quad (3)$$

⁹ In this setting, observable and unobservable characteristics are a function of amenities, and can be inverted as a function of observed characteristics (i.e., group average observables).

¹⁰ This multinomial model of utility is common in urban economics (e.g. [Kline and Moretti, 2014](#); [Diamond and Gaubert, 2021](#)).

Individuals choose where to live to maximize utility over all locations, taking the amenities, prices, and individual characteristics as given.

Altonji and Mansfield (2018) make five modeling assumptions: First, preferences are given by equation (3). Second, prices and amenities in each location are viewed as exogenous to the individual when choosing their location. Third, ε_{si} has mean zero and is independent of observables, unobservables, and amenities for all locations. Fourth, the expectation of X_i conditional on W_i and the expectation of X_i^U are both linear in W_i . Fifth, there are more observables than unobservables. The last assumption is called a rank condition and requires that the set of location-specific factors that drive sorting on *observable* covariates contains *all* location-specific factors that drive sorting based on *unobservable* covariates.¹¹ The intuition for this approach is that the regression will first “invert” the observed group-level averages to location-specific amenities from the fourth assumption above. Then these amenities can be linearly projected to group averages of the unobservable covariates using the fourth and fifth assumption. Hence, controlling for group averages of observed characteristics is sufficient to control for the sorting based unobserved characteristics.¹²

Taking this approach, we rewrite equation (1) to include group-level averages of the individual characteristics in each location. In particular, we estimate:

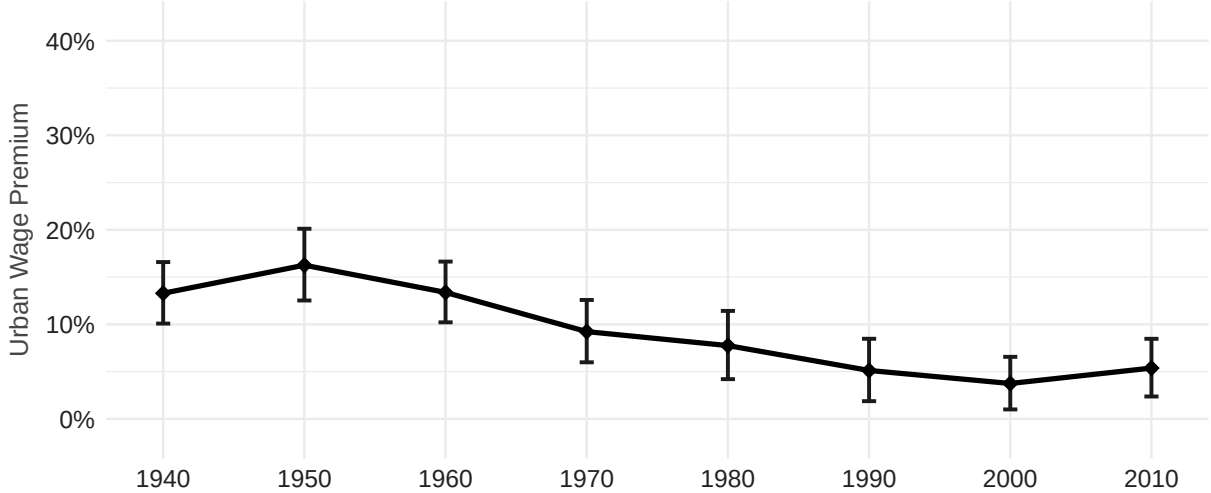
$$\log wage_i = \beta_0 + \beta_1 urban_i + X_i \delta + \bar{X}_{c(i)} \gamma + \varepsilon_i, \quad (4)$$

where $\bar{X}_{c(i)}$ are the location-specific average of the individual characteristics in X_i and addi-

¹¹ Diamond (2016) provides evidence using a principal component analysis on a very detailed set of amenities and documents that the effective rank of amenities is around five (with three components explaining most of the variation). In the context of this paper, the key identifying assumption is that our group-average variables have a rank larger than five. Empirically, we run principal components analysis and find more than five components that explain more than 5 percent of the total variance across years. This provides support for using the approach in Altonji and Mansfield (2018) to address sorting.

¹² The above “inversion” is only possible if the mapping between group averages and location-specific amenities is one-to-one. If there are more amenities than group-average variables, then a unique inverse will not exist. Hence, the rank condition is necessary.

Figure 2: The Urban Wage Premium Controlling for Individual Characteristics and Sorting Using Group-Level Averages, 1940-2010



Notes: This figure plots the coefficient for survey-year level estimates of the urban wage premium following the methodology in Section 4. We estimate equation (4) with the individual-level and group-average controls. These point estimates serve as our estimate for the causal urban wage premium.

tional location-invariant characteristics.¹³ In this case, β_1 now captures the urban premium after controlling for sorting. One concern with the approach is that it assumes the exogeneity of local housing prices. As an alternative approach, we also consider a robustness check that uses the individual wage net of a measure of local housing costs as the dependent variable in equation (4).

We present our main estimates of equation (4) for each year are plotted in Figure 2. These estimates make clear that using group-level average of individual characteristics to control for sorting decreases the urban wage premium significantly. Relative to the raw urban-nonurban wage differential, our estimates suggest that individual characteristics and sorting explain approximately 75 percent of the wage differential between urban and nonurban areas after 1980 and approximately half in earlier decades.

¹³ We include as group-averages the share of white workers (the share of non-white workers drops out); share of residents in all but one of the age groups; share that are veterans; share that are married and together, married and apart, and single; and share with less than a high school degree, share with a high school degree, share with some college, share with BA degree, and share with advanced education. Additionally, we include a measure of market access calculated as in Jaworski and Kitchens (2019).

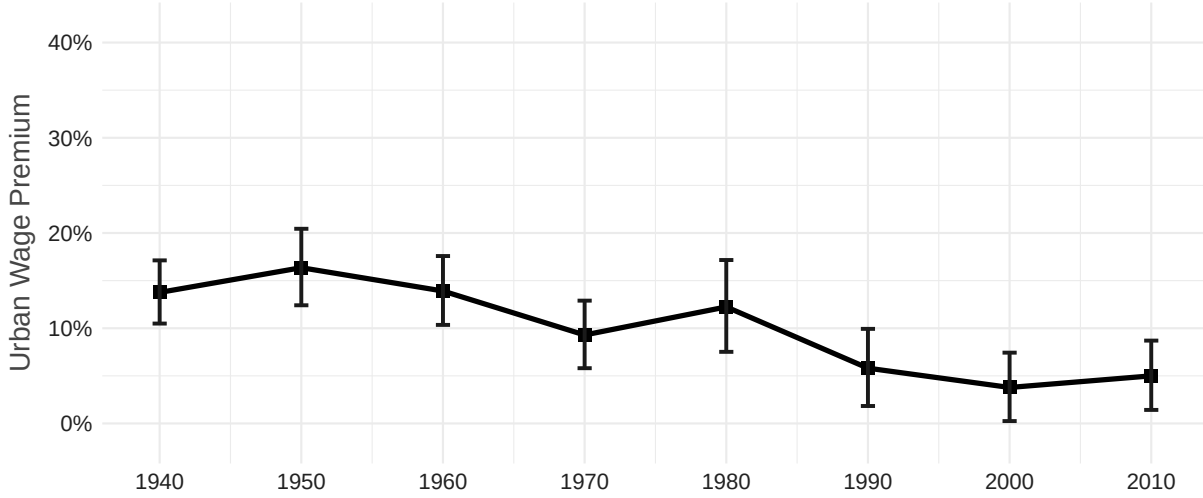
Focusing on the period between 1980 and 2010, when our estimates overlap those reported in the existing literature, we find an urban wage premium of approximately 5 percent in each year. Our estimates closely match those reported in [Glaeser and Maré \(2001\)](#), who report a premium of 4.5 percent in the densest metropolitan areas. Our estimates also mirror those reported by [Yankow \(2006\)](#), whose fixed effects estimates range from 4.9 to 5.4 percent. Our estimates are also within the range of those reported in [Baum-Snow and Pavan \(2012\)](#). The overlap of our estimates for the decades since 1980 and previous estimates provides support for our approach using group-level averages to address the potential role of sorting.

Next, we turn to the period from 1940 to 1980, where the previous literature has yet to provide estimates of the urban wage premium. In 1940, we find that the urban wage premium was approximately 13 percent. Given a raw differential of 31 percent, sorting and individual characteristics explain about three-fifths of the urban-nonurban wage gap. Between 1940 and 1970, we estimate that the urban wage premium fell substantially. The premium stayed high at 16 percent in 1950 and 13 percent in 1960 and then began to decline to about 5 percent starting in the 1990s.

The estimates for the period from 1940 to 1970 are striking. Our estimates for the earlier period are more than twice the size of those reported in the literature for the period between 1980 and 2020. The recent literature has sought to explain the widening urban-nonurban wage gap since 1970, largely focusing on the differential return to skill in urban areas ([Ganong and Shoag, 2017](#); [Baum-Snow, Freedman and Pavan, 2018](#); [Dauth, Findeisen, Moretti and Suedekum, 2018](#); [Autor, 2019](#)). For example, [Baum-Snow et al. \(2018\)](#) and [Autor \(2019\)](#) highlight that the returns to education in urban areas increased starting in 2000 due to agglomeration forces that are skill-biased.

Our estimates suggest that earlier periods of rapid change affected the wage differential between urban and nonurban areas. We show that sorting became increasingly important in postwar period, which is earlier than previously documented. Broadly, these results reinforce the focus of [Autor \(2019\)](#) and ? on the changing number and composition of occupations available for low-skill workers in this period. For example, our estimates are consistent with the wave of factory automation in the 1950s and early 1960s.

Figure 3: The Urban Wage Premium Using the 1940 Definition of MSAs, 1940-2010



Notes: This figure plots the coefficient for survey-year level estimates of the urban wage premium following the methodology in Section 4. We estimate equation (4) with the individual-level and group-average controls using only the subsample of 1940 MSAs. Individuals living in MSAs that were added after 1940 are dropped from the sample.

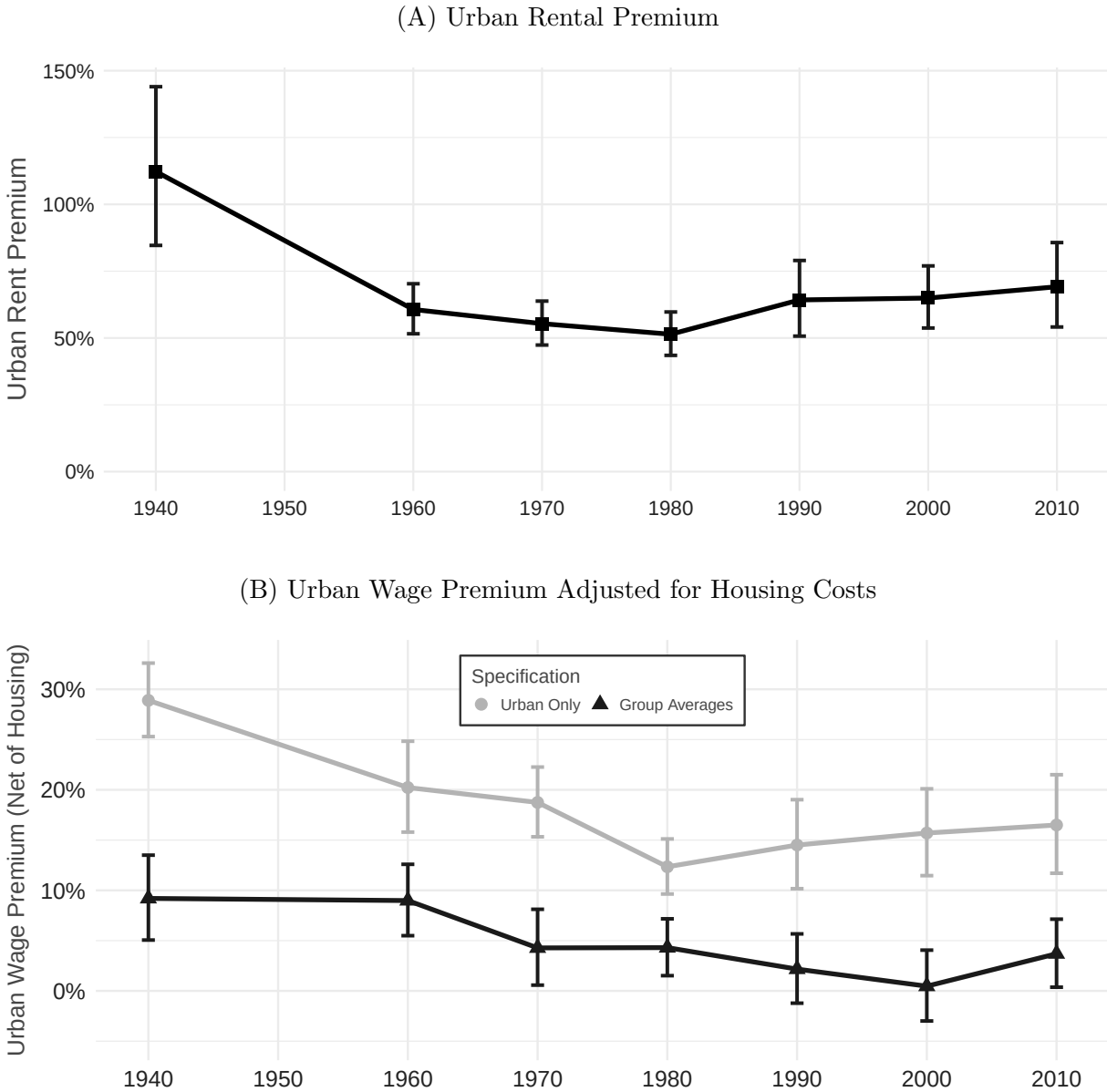
Over time, the number of MSAs grows as new cities increase in size. This raises a concern that our estimates may reflect a change in the composition of MSAs. To rule this out, we re-estimate equation (4) with a fixed sample of MSAs based on their 1940 designations and drop observations from MSAs designated after 1940 from the sample. In Figure 3 we report the estimates holding the set of MSAs constant. The results are similar to our baseline estimates, suggesting that the changing composition of cities does not bias our estimates.

Urban Wage Premium Adjusted for Housing Costs

So far we have assumed that housing prices are exogenous to individual location decisions. However, canonical urban models suggest that urban rents will adjust in response to higher wages and partially absorb the wage premium, such that utility is equalized across space. In Panel A of Figure 4 we document that urban rents were significantly higher than rents outside of MSAs. For example, between 1960 and 2010, the average rent in an MSA was between 50 and 70 percent higher than non-MSA regions.

To ensure that the urban wage premium we estimate is not driven by rent differentials,

Figure 4: Urban Rental and Housing Cost-Adjusted Wage Premium, 1940-2010



Notes: Panel A plots the coefficient for survey-year level estimates of the urban rental premium following the methodology in Section 4 using housing rents as the dependent variable. Panel B plots the coefficient for survey-year level estimates of the urban wage premium following the methodology in Section 4 using the individual wage adjusted for housing costs as the dependent variable. All specifications include individual-level controls and group-level averages.

we re-estimate equation (4) using a measure of income that accounts for location-specific differences in housing costs. Specifically, we calculate the average rent for each MSA and non-urban area in a given year and subtract it from our measure of annual income.¹⁴ Panel B of Figure 4 reports the results. Relative to our baseline estimates, accounting for differences in housing costs results in a smaller urban wage premium. However, the pattern of the estimates is similar. Between 1940 and 1970 the urban wage premium (adjusted for housing costs) decreases, remains low through 2000, and modestly increases back to the level in 1970.

5 Heterogeneity by City Size, Region, Income, and Education

Up to this point, we have documented the increasing role of sorting in explaining the difference between urban and non-urban wages over time. Importantly, this pattern may differ substantially throughout the city-size distribution, across regions, or for workers with different levels of education. In this section, we examine the role of these factors.

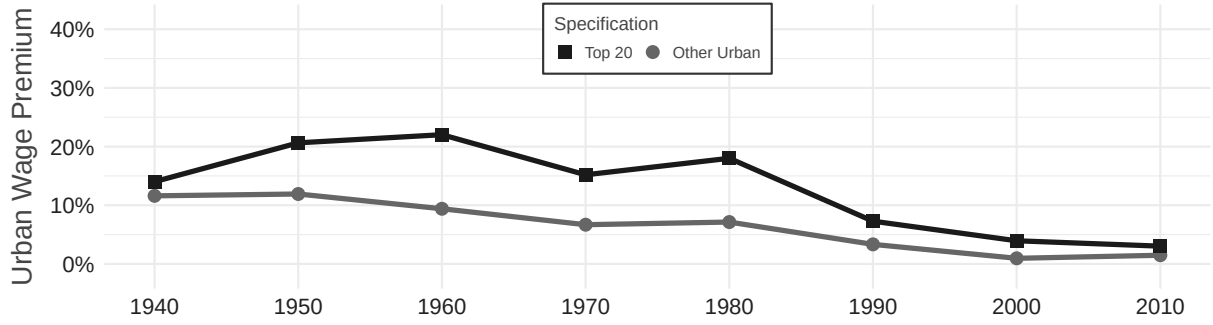
Recent literature highlights that the urban wage premium is increasing in city size (e.g. Baum-Snow and Pavan, 2012). To test for heterogeneity in the urban wage premium throughout the city-size distribution, we re-estimate equation (4) but allow cities in the 20 most populated MSAs as of 1940 to have a differential effect. Panel A of Figure 5 plots the estimated urban wage premium for the twenty largest MSAs (square markers) versus the remaining MSAs (round markers). First, we document that the urban wage premium is higher in the largest 20 MSAs relative to smaller MSAs in all years, consistent with the existing literature. While the largest MSAs generate larger wage premia, the returns to the largest MSAs have declined relative to smaller MSAs in the last 30 years.

Previously, we highlighted how the geography of the highest-wage cities has varied over time. In the 1940s, the largest premiums were in Rust Belt cities like Flint, MI and Rochester, NY. Today, the pattern has shifted towards Sunbelt cities and technology hubs. Given the shift in the location of work, the relative importance of sorting and agglomeration may also differ across space over the last 80 years. To explore how the relative importance of sorting

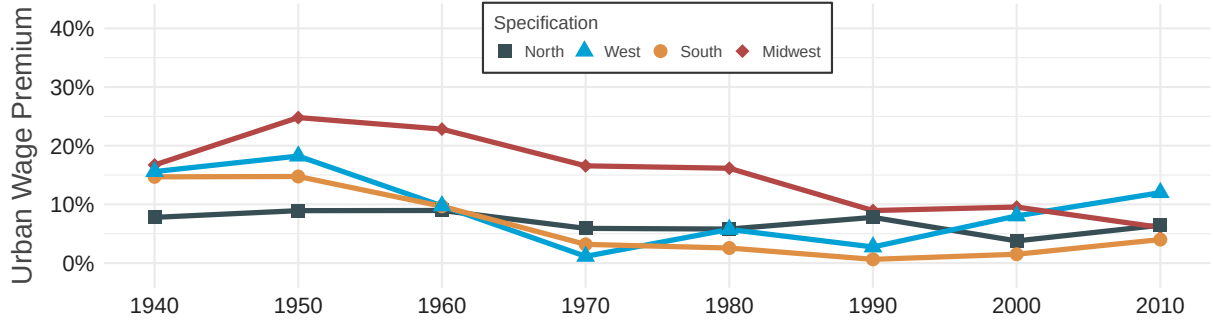
¹⁴ We do not include 1950 since individual-level data on gross rents are not available. For 1940, we drop reported rents larger than 150 (in 1940 dollars). We use the average rent in an MSA to abstract away from heterogeneous taste for housing quality.

Figure 5: Heterogeneity of the Urban Wage Premium, 1940-2010

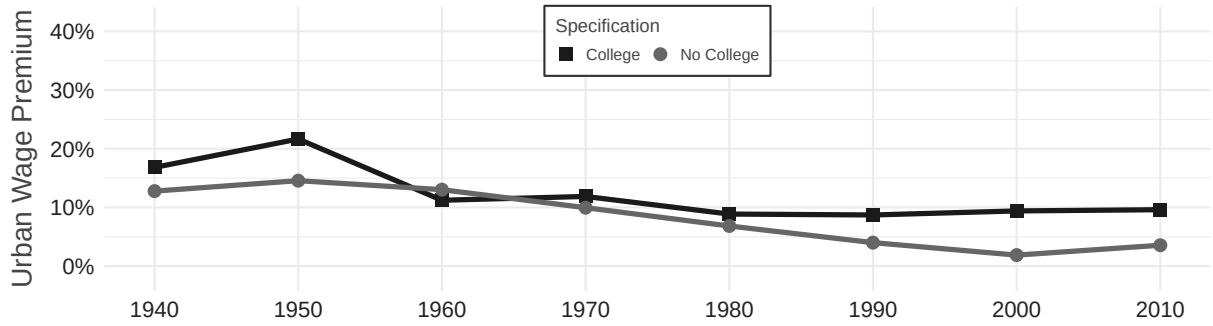
(A) By Twenty Largest versus Other Cities



(B) By Census Regions



(C) By College versus No College Degree



Notes: This figure plots the coefficient for survey-year level estimates of the urban wage premium from equation (1) including the individual-level and group-average controls described in footnote 13 while interacting the urban indicator with three different variables. Panel A interacts the urban indicator with an indicator for the top 20 most populous MSA in 1940. Panel B interacts the urban indicator with indicators for each census region. Panel C reports estimates of equation (1) run on separate samples of college-educated and non-college-educated workers with the group-averages taken separately by sample.

varies across geography, we estimate specifications to allow for differences in each census region (i.e., Northeast, South, Midwest, and West). Panel B of Figure 5 plots the estimates. The Midwest has experienced a large decline in the share of the premium attributable to agglomeration, while agglomeration forces have become relatively more important in the West since 1970.

As automation and technology have increasingly entered the workplace, the returns to cities may also vary over time for workers with different levels of education. For example, Gould (2007) shows that the urban wage premium is significantly larger for white-collar workers. To explore how agglomeration forces have varied by education group, we estimate separate specifications for college-educated workers and those with less than a college degree.¹⁵ Panel C of Figure 5 plots our estimates by college degree status. Throughout the period, we generally estimate larger wage premia for college-educated workers, on the order of magnitude of 3-5 percentage points. Comparing our estimates to Gould (2007), we similarly find a premium of around 10 percent for college-educated workers in the 1980s.

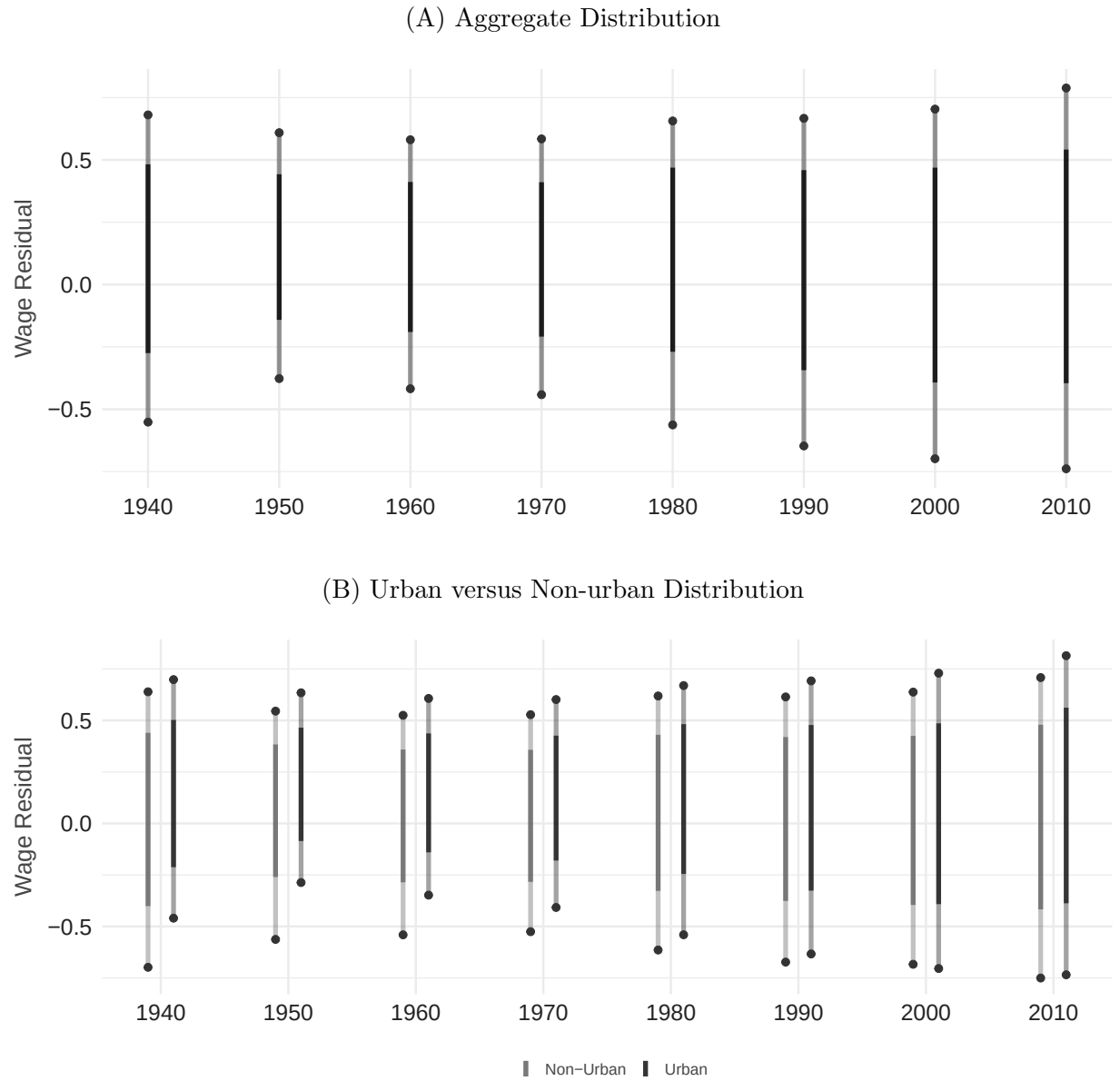
In the period after World War II, there were substantial changes in wage inequality. Goldin and Margo (1992) discuss the narrowing of the wage distribution in what they label the “Great Compression” between 1940 and 1960. They highlight the rapid increase in the supply increase of educated workers during a period of relatively stagnant demand growth. Recently, Vickers and Ziebarth (2022) discuss the role of the National War Labor Board on wages in various regions throughout the country and show that wage-setting policies had a meaningful impact on local wage distributions.

To explore whether we observe a compression and then widening of the wage distribution in our setting, we take the residuals from (4) which reflect individual earnings after controlling for sorting. The ninetieth-tenth percentiles summarize the distribution of the wage residuals for each decade in Panel A of Figure 6. Similar to Goldin and Margo (1992), we find evidence consistent with a narrowing of the wage distribution through 1960. Afterward, the wage distribution widens considerably through 2010.

In Panel B of Figure 6, we plot the distribution of residuals for urban and non-urban

¹⁵ We calculate the group averages separately by group to allow the sorting mechanisms to differ by group.

Figure 6: Distribution of Wage Residuals, 1940-2020



Notes: This figure takes the residual of wages after regressing on the full-specification and adds back in the causal urban wage premium. Each panel plots points for the 90th-10th percentiles of the wage residuals and shades darker the 80th-20th percentiles. Panel A does this for the entire distribution of wages following [Goldin and Margo \(1992\)](#). Panel B does this separately for urban and non-urban workers.

workers. In 1940, urban and non-urban workers had a large left tail in the wage distribution, reflecting many workers who earned very low (residualized) weekly earnings. Between 1940 and 1960 this left-tail disappeared, which provides some evidence that a rising floor potentially drove the Great Compression on non-urban wages.¹⁶ [Goldin and Margo \(1992\)](#) find similar evidence that increases in the distribution for low-wage workers drove the Great Compression. Our estimates suggest that the narrowing of the wage distribution occurred in urban and non-urban areas.

6 Conclusion

Wages in cities are much higher than in non-urban areas. Understanding why this gap exists and, in particular, the relative role of sorting versus agglomeration over time is an important topic in urban economics. In this paper, we quantify the urban wage premium in each decade between 1940 to 2010. Importantly, we show how to apply the methodology of [Altonji and Mansfield \(2018\)](#), which enables us to use data from annual cross-sections to account for sorting between urban and rural locations. Thus, we provide new estimates of the urban wage premium for the period from 1940 to 1980, while generating estimates for the period from 1980 to 2010 that are similar to those in the existing literature.

Our results highlight the increasing relative importance of sorting over the second half of the twentieth century and into the first part of the twenty-first century. By 2010, the relative strength of agglomeration forces was less than half of the agglomeration forces we estimate in 1940. In addition, we find that this pattern is not driven by changes in the composition of cities in our sample over time or differences in housing costs across cities. We also show that changes in the urban wage premium and the role of sorting are not due to changes in the returns to city size, regions, level of education, or the structure of the wage distribution across urban and non-urban areas. Overall, our findings are consistent with a greater role for consumer amenities in explaining the demand for urban living as of 2010.

¹⁶ The overall compression and the narrowing of the left-tail are also apparent when using 95th and 5th percentiles.

References

- Ahlfeldt, Gabriel M, Thilo Albers, and Kristian Behrens, “A granular spatial model,” 2022.
- Akcigit, Ufuk, Salomé Baslandze, and Stefanie Stantcheva, “Taxation and the International Mobility of Inventors,” *American Economic Review*, 2016, *106* (10), 2930–81.
- Altonji, Joseph G. and Richard K. Mansfield, “Estimating Group Effects Using Averages of Observables to Control for Sorting on Unobservables: School and Neighborhood Effects,” *American Economic Review*, 2018, *108* (10), 2902–46.
- Autor, David H., “Work of the Past, Work of the Future,” *AEA Papers and Proceedings*, May 2019, *109*, 1–32.
- Baum-Snow, Nathaniel and Daniel Hartley, “Accounting for Central Neighborhood Change, 1980–2010,” *Journal of Urban Economics*, 2020, *117*, 103228.
- and Ronni Pavan, “Understanding the City Size Wage Gap,” *Review of Economic Studies*, 2012, *79* (1), 88–127.
- , Matthew Freedman, and Ronni Pavan, “Why Has Urban Inequality Increased?,” *American Economic Journal: Applied Economics*, 2018, *10* (4), 1–42.
- Behrens, Kristian, Gilles Duranton, and Frédéric Robert-Nicoud, “Productive Cities: Sorting, Selection, and Agglomeration,” *Journal of Political Economy*, 2014, *122* (3), 507–553.
- Boustan, Leah Platt, Devin Michelle Bunten, and Owen Hearey, “Urbanization in the United States, 1800-2000,” Working Paper 19041, National Bureau of Economic Research May 2013.
- Combes, Pierre-Philippe, Gilles Duranton, and Laurent Gobillon, “Spatial Wage Disparities: Sorting Matters!,” *Journal of Urban Economics*, 2008, *63* (2), 723–742.
- Dauth, Wolfgang, Sebastian Findeisen, Enrico Moretti, and Jens Suedekum, “Matching in Cities,” *Journal of the European Economic Association*, 2018.
- , —, —, and —, “Matching in cities,” *Journal of the European Economic Association*, 2022, *20* (4), 1478–1521.
- Diamond, Rebecca, “The Determinants and Welfare Implications of US Workers’ Diverging Location Choices by Skill: 1980–2000,” *American Economic Review*, Mar 2016, *106* (3), 479–524.
- and Cecile Gaubert, “Spatial Sorting and Inequality,” *Working Paper*, 2021.
- Dingel, Jonathan I and Felix Tintelnot, “Spatial economics for granular settings,” Technical Report, National Bureau of Economic Research 2020.
- Dumais, Guy, Glenn Ellison, and Edward L. Glaeser, “Geographic Concentration as a Dynamic Process,” *Review of Economics and Statistics*, 2002, *84* (2), 193–204.

- Frings, Hanna and Rebecca Kamb, “The Relative Importance of Portable and Non-Portable Agglomeration Effects for the Urban Wage Premium,” *Regional Science and Urban Economics*, 2022, *95*, 103786.
- Ganong, Peter and Daniel Shoag, “Why Has Regional Income Convergence in the US Eeclined?,” *Journal of Urban Economics*, 2017, *102*, 76–90.
- Glaeser, Edward L. and David C. Maré, “Cities and Skills,” *Journal of Labor Economics*, Apr 2001, *19* (2), 316–342.
- and Giacomo A.M. Ponzetto, “Did the Death of Distance Hurt Detroit and Help New York?,” *Agglomeration Economics*, 2010, *303*, 303–05.
- and Jesse M. Shapiro, “Urban Growth in the 1990s: Is City Living Back?,” *Journal of Regional Science*, 2003, *43* (1), 139–165.
- and Joshua D. Gottlieb, “The Wealth of Cities: Agglomeration Economies and Spatial Equilibrium in the United States,” *Journal of Economic Literature*, 2009, *47* (4), 983–1028.
- Goldin, Claudia and Robert A Margo, “The great compression: The wage structure in the United States at mid-century,” *The Quarterly Journal of Economics*, 1992, *107* (1), 1–34.
- Gould, Eric D., “Cities, Workers, and Wages: A Structural Analysis of the Urban Wage Premium,” *Review of Economic Studies*, 2007, *74* (2), 477–506.
- Jaworski, Taylor and Carl T. Kitchens, “National Policy for Regional Development: Historical Evidence from Appalachian Highways,” *Review of Economics and Statistics*, 2019, *101* (5), 777–790.
- Kline, Patrick and Enrico Moretti, “People, Places, and Public Policy: Some Simple Welfare Economics of Local Economic Development Programs,” *Annual Review of Economics*, Aug 2014, *6* (1), 629–662.
- Moretti, Enrico, “Local Labor Markets,” in “Handbook of Labor Economics,” Vol. 4 2011, pp. 1237–1313.
- , “Real Wage Inequality,” *American Economic Journal: Applied Economics*, 2013, *5* (1), 65–103.
- and Daniel J. Wilson, “The Effect of State Taxes on the Geographical Location of Top Earners: Evidence from Star Scientists,” *American Economic Review*, 2017, *107* (7), 1858–1903.
- Puga, Diego, “The Magnitude and Causes of Agglomeration Economies,” *Journal of Regional Science*, 2010, *50* (1), 203–219.
- Rosenthal, Stuart S. and William C. Strange, “Evidence on the Nature and Sources of Agglomeration Economies,” in “Handbook of Regional and Urban Economics,” Vol. 4, Elsevier, 2004, pp. 2119–2171.

- Ruggles, Steven, Catherine A. Fitch, Ronald Goeken, J. David Hacker, Matt A. Nelson, Evan Roberts, Megan Schouweiler, and Matthew Sobek, “IPUMS Ancestry Full Count Data: Version 3.0 [dataset],” in “,” IPUMS, 2022.
- , Sarah Flood, Ronald Goeken, Megan Schouweiler, and Matthew Sobek, “IPUMS USA: Version 12.0 [dataset],” in “,” IPUMS, 2022.
- Schoefer, Benjamin and Oren Ziv, “Productivity, Place, and Plants,” *Review of Economics and Statistics*, 2022, pp. 1–46.
- Shapiro, Jesse M., “Smart Cities: Quality of Life, Productivity, and the Growth Effects of Human Capital,” *Review of Economics and Statistics*, 2006, *88* (2), 324–335.
- Su, Yichen, “The Rising Value of Time and the Origin of Urban Gentrification,” *American Economic Journal: Economic Policy*, 2022, *14* (1), 402–39.
- Verginer, Luca and Massimo Riccaboni, “Talent Goes to Global Cities: The world Network of Scientists’ Mobility,” *Research policy*, 2021, *50* (1), 104127.
- Yankow, Jeffrey, “Why Do Cities Pay More? An Empirical Examination of Some Competing Theories of the Urban Wage Premium,” *Journal of Urban Economics*, 2006, *60* (2), 139–161.

Table 3: Urban Wage Premium, 1940-2010

	log(Weekly Wage)							
	1940	1950	1960	1970	1980	1990	2000	2010
<i>Panel A: Raw Wage Premium</i>								
Urban = 1	0.3140*** (0.0176)	0.2208*** (0.0131)	0.2295*** (0.0135)	0.2109*** (0.0119)	0.1692*** (0.0110)	0.1943*** (0.0157)	0.2018*** (0.0144)	0.1921*** (0.0155)
<i>Panel B: Individual Controls</i>								
Urban = 1	0.2797*** (0.0168)	0.2012*** (0.0125)	0.2045*** (0.0123)	0.1833*** (0.0119)	0.1334*** (0.0116)	0.1615*** (0.0162)	0.1571*** (0.0125)	0.1436*** (0.0128)
<i>Panel C: Individual Controls & Group Averages</i>								
Urban = 1	0.1248*** (0.0146)	0.1506*** (0.0167)	0.1256*** (0.0145)	0.0883*** (0.0154)	0.0747*** (0.0171)	0.0500*** (0.0160)	0.0368*** (0.0137)	0.0524*** (0.0148)
Observations	2,540,851	68,457	1,364,970	280,338	1,910,440	2,262,149	2,583,088	2,720,062

Notes. This table reports coefficient for survey-year level estimates of the urban wage premium from equation (1) with three different level of controls. The first section includes no controls and represents the urban wage premium observed in the data. The second section includes controls for age in 5-year bins, education levels, and an indicator for being White. The final section includes group-average variables following the methodology in Section 4. These point estimates serve as our estimate for the causal urban wage premium.

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1. Standard errors are clustered by MSA/CBSA.